

# BANA 4373: Advanced Business Analytics for Economics and Business

## Lecture 1: Data in Economics, Business, and Policy

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# Why This Course Exists

- Economics has become a **data-driven discipline**
- Policy, business, and research decisions rely on empirical evidence
- This course focuses on **using data to answer economic questions**

# How Economists Use Data

- Economists care about:
  - Decisions
  - Incentives
  - Trade-offs
- Data is a **tool**, not the final goal

# Economics vs. Other Data Fields

**Statistics**

Estimation, inference, uncertainty

**Computer Science**

Algorithms, efficiency, scalability

**Economics (this course)**

Decisions, policy, counterfactuals

# Two Fundamental Questions

## Prediction

*What will happen?*

- Forecast unemployment
- Predict sales or prices

*This is the main goal of many machine learning models.*

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## Explanation / Causality

*Why did it happen?*

- Did a policy cause the change?
- What would have happened otherwise?

*This is central to economic and policy analysis.*

# Where These Appear in BANA 4373 (Course Timeline)

- **Prediction** shows up throughout the course, especially when we focus on:
  - **Weeks 4–5:** Visualization & dashboards (communicating patterns)
  - **Weeks 10–11:** Machine learning (forecasting, classification)
  - **Week 12:** Deployment (turning predictions into tools)
- **Explanation / Causality** is central when we focus on:
  - **Week 6:** Applied regression (interpreting relationships carefully)
  - **Week 7:** Causal inference (DiD, policy evaluation intuition)
- **Projects (Weeks 13–15):** You will choose an applied question that is primarily:
  - *Prediction-focused* or *Causality-focused* (or a mix)

*Different tools for different questions — the key skill is knowing which one you need.*

# Prediction vs. Causation

- Correlation does **not** imply causation
- Machine learning is great for prediction
- Policy analysis requires causal thinking

*This distinction will appear throughout the course.*

# Types of Data Economists Use

- **Cross-sectional data**
  - Many units, one point in time
- **Time series data**
  - One unit, many points in time
- **Panel data**
  - Many units, many points in time

# Examples of Economic Data Structures

- **Cross-sectional example:**

- Household income across individuals in 2023
- Question: Who earns more and why?

- **Time series example:**

- U.S. unemployment rate, monthly from 1960–2025
- Question: What is happening over time?

- **Panel data example:**

- County-level unemployment rates, 2000–2023
- Question: How do places change over time?

# Real Data Sources by Type

- **Cross-sectional data**

- American Community Survey (ACS)
- [census.gov/programs-surveys/acs](https://www.census.gov/programs-surveys/acs)

- **Time series data**

- FRED: Unemployment Rate, CPI, GDP
- [fred.stlouisfed.org](https://fred.stlouisfed.org)

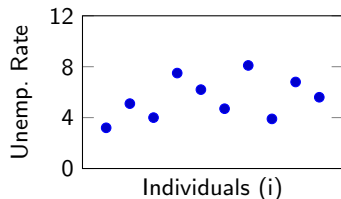
- **Panel data**

- BLS Local Area Unemployment Statistics (LAUS)
- [bls.gov/lau](https://www.bls.gov/lau)

# Same Variable, Three Data Structures

## Cross-Section (one time)

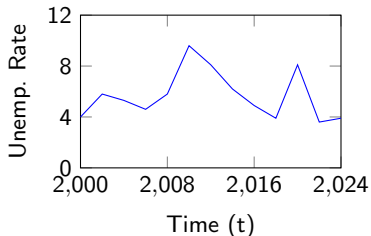
2024 snapshot



Many units, one time  
(compare people/places)

## Time Series (one unit)

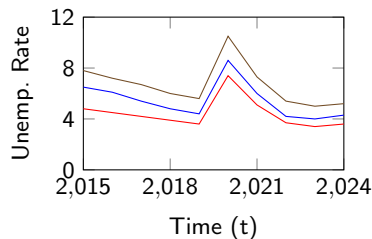
U.S. over time



One unit, many times  
(trend/forecast)

## Panel (many units)

3 counties (example)



Many units, many times  
(changes + comparisons)

**Same variable, different structure → different questions you can answer.**

# Why Data Structure Matters

- Different data structures answer different questions
- Using the wrong structure can lead to:
  - Incorrect conclusions
  - Bad policy recommendations
  - Misleading business decisions
- Economists must choose the **right data for the question**

# The Economic Data Ecosystem

- **FRED (Federal Reserve Economic Data)**  
[fred.stlouisfed.org](https://fred.stlouisfed.org)
- **BLS (Bureau of Labor Statistics)**  
[bls.gov](https://bls.gov)
- **BEA (Bureau of Economic Analysis)**  
[bea.gov](https://bea.gov)
- **U.S. Census / ACS**  
[census.gov](https://census.gov)
- **World Bank Data**  
[data.worldbank.org](https://data.worldbank.org)

# APIs and Programmatic Access

## What is an API?

- An **API (Application Programming Interface)** is a way for a computer program to **ask another computer for data**.
- Instead of downloading files manually, your code sends a request and receives data automatically.

## Why economists use APIs:

- Always up-to-date data
- Reproducible analysis
- Easy access to large datasets

## Examples we will use:

- FRED API: [fred.stlouisfed.org/docs/api](https://fred.stlouisfed.org/docs/api)
- Census API: [census.gov/data/developers](https://census.gov/data/developers)
- World Bank API: World Bank API Docs

*Think of an API as a waiter: you place an order, the kitchen sends back exactly what you asked for.*

# Reading Data Like an Economist

Before running code, always ask:

- Who collected this data?
- Why was it collected?
- What incentives shaped the data?
- What decisions could it inform?

# Data Dictionary: The Map of Your Dataset

- A **data dictionary** explains **what each variable means**.
- It typically includes:
  - Variable name
  - Definition
  - Units and scale
  - Time frequency
  - Valid values or codes

## Example: FRED Unemployment Rate (UNRATE)

- **Variable name:** UNRATE
- **Definition:** Percentage of the labor force that is unemployed
- **Units:** Percent
- **Frequency:** Monthly
- **Source:** Bureau of Labor Statistics (via FRED)

*If you don't know the definition or units, your interpretation will be wrong.*

# Metadata: The Context Behind the Numbers

- **Metadata** is “data about the data.”
- It provides the context needed to evaluate and use a dataset correctly.

## Example: What does “Unemployed” mean?

- Collected by: **Bureau of Labor Statistics**
- Data collection method: **Current Population Survey (CPS)**
- Population: **Civilian, non-institutional population (age 16+)**
- Definition of unemployed:
  - Not working
  - Actively looked for work in last 4 weeks
  - Available to work
- Important limitation:
  - People who stopped searching are **not counted as unemployed**

*This definition matters for policy debates and economic conclusions.*

# In-Class Example: FRED

- Unemployment rate
- Inflation (CPI)
- What happened in 2008? 2020?

Explore live data on FRED

# Where This Course Fits

- Only economics course required for all Data Science tracks
- Bridges statistics, computer science, and economics
- Focus on **applied tools + interpretation**

# What You Will Learn

- Data cleaning and pipelines
- Visualization and dashboards
- Regression and causal reasoning
- Machine learning (applied)
- Communicating results

# Looking Ahead

- Next class: data cleaning and reproducibility
- Real-world data is messy — we embrace that

# Key Takeaways

- Economics is a decision science
- Data without context is dangerous
- Tools matter — judgment matters more